

Q2: human eye

1. Cross-section of human eye

2. functions of the main components of human eye pertaining to visual system

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Cornea: Transparent tissue that protects the eye

Sclera: Opaque membrane of optic globe

Choroid: Contains blood vessels. Extremely pigmented to protect backscatter inside the optic globe.

Ciliary body: Adjust focal length of optical lens

Iris: control the amount of light entering the optic globe

Lens: Concentric layers of fibrous cells. Absorbs 8% of visible spectrum with high absorption at short wavelength. Absorption in Ultraviolet or Infra red light may damage eye.

Retina: Consists of two receptors : cones and rods.

Cone Receptors: Fewer in number and concentrated in the **fovea**. They provide high color sensitivity and fine detail because each cone connects to a single nerve.

Rod Receptors: Highly numerous and distributed across the eye. They are sensitive to **low light** but provide lower resolution since multiple rods share a single nerve connection.

Cross-section of human eye

